Project Management & Engineering (PME)
Ma'aden Engineering Manual

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REVISION INDEX

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1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for the Earthworks for all Ma'aden projects.

This document is intended to be used in conjunction with the referenced documents listed herein.

1.2 Purpose

The purpose of this specification is to set the minimum engineering standard for Earthworks for all Ma'aden projects.

2. KEY TERMS AND DEFINITIONS

Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform Work or cause Work to be performed.
Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
Site	Geographical location where Services are to be performed.
Work	The entire work and Services and all things which the Contractor is or may be required to execute, supply and do under the Contract having regard to the Scope of Project, or to cause to be executed, supplied or done, for the purposes of constructing and Commissioning the Plant.

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All Work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any

way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All material, equipment and systems shall conform to the following.

3.2.1 Local Codes and Standards

SBC303 Soil and Foundations Requirements

3.2.2 International Codes and Standards

3.2.2.1 American Concrete Institute (ACI)

ACI 229 Controlled Low Strength Material (CLSM)

3.2.2.2 American Society for Testing and Materials (ASTM)

ASTM C136 Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates

ASTM D422 Standard Test Method for Particle-Size Analysis of Soils

ASTM D698 Standard Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort 12,400 ft-lbf/ft3 (600 kN-m/m^3) Referred to as a "Standard Proctor Compaction Test"

ASTM D1140 Standard Test Method for Amount of Material in Soils Finer than the No. 200 (75 □m) Sieve

ASTM D1241 Standard Specification for Materials for Soil-Aggregate Sub-Base, Base and Surface Courses

ASTM D1556 Standard Test Method for Density and Unit Weight of Soil in Place by the Sand-Cone Method

ASTM D1557 Test Method for Laboratory Compaction Characteristics of Soil Using Modified Effort 56,000 ft-lbf/ft3 (2700 kN-m/m^3) Referred to as a "Modified Proctor Compaction Test"

ASTM D2167 Standard Test Method for Density and Unit Weight of Soil in Place by the Rubber Balloon Method

ASTM D2487 Classification of Soils for Engineering Purposes (Unified Soil Classification System)

ASTM D2922 Standard Test Methods for Density of Soils and Soil-Aggregate in Place by Nuclear Methods (Shallow Depth)

ASTM D2937	Standard Test Method for Density of Soil in Place by the Drive-Cylinder Method
ASTM D2940	Standard Specification for Graded Aggregate Material for Bases or Sub-Bases for Highways or Airport
ASTM D3282	Standard Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes
ASTM D3740	Standard Practice for Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock Used in Engineering Design and Construction
ASTM D4253	Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table
ASTM D4254	Standard Test Method for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density
ASTM D4318	Standard Test Method for Liquid Limit, Plastic Limit and Plasticity Index of Soils
ASTM D5971	Standard Practice for Sampling Freshly Mixed Controlled Low-Strength Material

3.2.2.3 Process Industry Practices (PIP)

PIP CVS02100	Site Preparation, Excavation, and Backfill Specification
PIP CVS02700	Underground Gravity Sewers Specification

3.2.3 Alternative Codes

Alternative codes may be used for Earthworks if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMEM-EG-CV-SPC-0003	Roadworks
MD-101-SMEM-EG-CV-SPC-0004	Stormwater Drainage and Gravity Sewers
MD-101-SMPM-PM-QA-PEI-0001	Contractor Quality Requirements

4. MATERIALS

4.1 General

Fill and backfill materials shall be from either on-site excavation, designated borrow areas or from off-site sources proposed by the Contractor and approved by the Employer. All proposed borrow sources shall be submitted with representative laboratory testing to include (at a minimum): particle size analysis, plasticity characteristics, and moisture-

density relationship (standard and/or modified proctor testing).

4.2 Definitions

Brackish Water	Water with Total Dissolved Solids (TDS) of 10,000 ppm or more.
Clean Sand	Soil meeting the following gradation and containing less than 3% weight of organic material or clay:

Sieve Size	Percent Passing
¼ inch (6.25 mm)	100
No. 10 (2.00 mm)	90 – 100
No. 200 (0.075 mm)	5 or less

Dune Sand	It Is wind-deposited sand consisting of relatively single-sized particles generally passing the No. 16 sieve (1.18 mm) and usually containing less than 10% soil particles passing the No. 200 sieve (0.074 mm).
Geomembrane	Thin sheet of polymeric material used as a liquid or vapor barrier. Refer to Specification MD-101-SMEM-EG-CV-SPC-0007.
Marl	It is a wide variety of calcareous sediments found in Saudi Arabia which may vary from clay to gravel sizes and often include cobble and boulder sized pieces.
Overbreak	The amount of ground unintentionally dislocated beyond the theoretical excavation line/profile.
Rock	Rock may refer to anything from boulder-size individual pieces to hard, intact bedrock and have various hardness and engineering properties. Classification of Rock by excavation methods is provided in Section 5.6.
Sabkha	Saline (generally coastal, but could also be inland) flats or deposits underlain by sand, silt or clay and often encrusted with salt.
Sabkha Brine	Saline water within the Sabkha and it is characterized by the high salinity where the salt concentrations are usually few times those of sea water. TDS greater that 35,000 ppm.
Select Fill	Select fill shall be composed only of inorganic material and shall have 100% passing the 75 mm (3") sieve and from 0 to 20% passing the No. 200 sieve. That portion of the material passing the No. 40 sieve shall have a maximum liquid limit of 40 and a maximum plasticity index of 20 per ASTM D4318. Liquid limit

and plastic limit tests are not required for select fill material with less than 15% passing the No. 200 sieve.

Structural Fill	Refer to Section 4.3.2.
Topsoil	Surface soil usually including the organic layer in which plants have nots of their roots. The thin, rich layer of soil where most of the nutrients for plant are found.

4.3 Fill and Backfill

4.3.1 General

Fill and backfill shall:

- Have a maximum particle size equal to 1/3rd of the compacted layer thickness.
- Not contain organic matter, chunks of highly plastic clay, contamination or other unsuitable material
- Not be gap graded or degradable
- Be as described in the geotechnical report (where applicable), and
- Not contain Rock or stones larger than 75 mm.

4.3.2 Structural Fill

Fill and backfill to be used as Structural Fill shall be gravel, sand or a mixture of these constituents, and shall have the following characteristics:

- Liquid limit (LL): 40 maximum, and
- Plasticity index (PI): 20 maximum.

4.3.2.1 Well Graded Sand

A well graded sand, by definition shall comprise Clean Sand with less than 5% fines (silt or clay size particles) passing the 75 □m sieve and more than 50% (by weight) passing the 2.36 mm sieve. The uniformity coefficient (C_u)[1] of the particle grading curve shall be greater than 6, and the coefficient of curvature (C_c)[2] of the particle grading curve shall range from more than 1 to less than 3.

4.3.2.2 Well Graded Gravel

This includes silty or clayey gravels (100% passing the 50 mm sieve, 50% or more of

[1] $C_u = \frac{D_{60}}{D_{10}}$

[2] $C_c = \frac{(D_{30})^2}{D_{10} \times D_{60}}$

Where D_{10}, D_{30} and D_{60} are the grain sizes of the particle expressed in millimeters at which 10%, 30% and 60% are finer.

coarse fraction retained on 4.75 mm sieve with 5% fines < 75 □m). The values for uniformity coefficient $(C_u)^1$ of the particle grading curve and the coefficient of curvature $(C_c)^2$ of the particle grading curve shall be $C_u \geq 4$, and $1 \leq C_C \leq 3$.

4.3.3 ***Controlled Low-Strength Material (CLSM)***

CLSM or "Flowable Fill" shall conform to ACI 229, unless otherwise specified. The mix design shall be proportioned to produce a minimum 28-day compressive strength between 1.0 and 2.1 MPa (about 140 to 300 psi), and shall be pre-tested using the actual raw materials. Sampling shall be done as per ASTM D5971.

4.3.4 ***Compaction Water***

Water used for compaction of fill material shall be reasonably free of salts, oil, acid, alkalis organic matter, or other deleterious substances. Use of water with a TDS in excess of 10,000 ppm such as brackish water is not allowed unless it is approved by the Employer except as indicated in Section 5.9.1.

4.4 Capping Materials

This material shall be well graded 0-20 mm crushed stone or well graded natural gravel complying with the requirements of sub-base as given in MD-101-SMEM-EG-CV-SPC-0003 – Roadworks.

4.5 Unsuitable Materials

Unsuitable materials include perishable or combustible materials, peat, Rocks or any buried debris, highly sensitive plastic clay and expansive clay that is not capable of meeting the requirements of this standard as to compaction and soil classification. This shall include materials which are difficult to compact or which have a liquid limit exceeding 35 as determined by ASTM D4318 and/or a plasticity index exceeding 12 as determined by ASTM D4318, and salt contaminated deposits (Sabkha).

5. EXECUTION

5.1 Site Examination and Set Out

The Contractor shall examine the Site and verify all existing levels, survey control points and the set out points shown on the drawings, before commencing the earthworks.

The earthworks shall be set out in accordance with the drawings.

All lines and levels shown on the drawings shall be confirmed on Site. The Contractor shall be solely responsible for taking all measures necessary to ensure the accurate location of new Work and its alignment with existing Work. The Employer will supply benchmarks.

5.2 Buried Services

The Contractor shall obtain all relevant underground service and utility drawings and also use underground pipe detectors prior to any digging.

The Contractor shall ascertain the position of all underground services and structures in the vicinity of the Work by hand digging or by other means approved by the Employer. If

applicable, the Contractor shall obtain a dig permit and vicinity permit from relative authority/s before starting excavations.

The Contractor shall ensure that the existing services are not damaged.

The Contractor shall notify the Employer immediately of any services encountered during the progress of the Works. These findings shall be clearly indicated on Site and noted in the As-Built' drawings.

5.3 Survey

A qualified surveyor shall perform the following survey works:

- Pre-construction survey
- Setting out the Works
- 'As constructed' survey. All drawings and information from the 'as constructed' survey shall be provided in the format as detailed in the specification.

5.4 Site Clearance

The Contractor shall give at least 48 hours notice of his intention to commence Site clearing. The Employer will, if applicable, nominate any trees, flora or other items which are required to remain intact within the area to be cleared. Within certain areas for clearing and grubbing, the Employer may nominate areas of partial clearing and grubbing or clearing only.

The Contractor shall not commence clearing any area until the Employer has approved that such area may be cleared.

Before removing topsoil, the Contractor shall clear the area of all debris and vegetation. Topsoil shall be removed to full depth as specified by the Employer or in the drawings.

Topsoil salvaged from within the limits of cut and fill shall be stored in stockpiles within the height limit listed below.

The classification of soil shall be determined as per ASTM D2487

All cleared areas shall be kept clear of re-growth for the duration of the Works.

Clearing and grubbing of the Site shall include the removal of all vegetation, deleterious matter such as tree stumps, roots, rubbish, building rubble and other debris within the area of the Work. It shall include the grubbing out of all stumps to a depth of 600 mm and all other roots, vegetation and boulders to a depth of 300 mm below the natural surface, the finished sub-grade surface (whichever is the lower) or as otherwise shown on the drawings. The waste matter shall be removed from the Site and disposed of as directed by the Employer to a spoil Site. (Refer to the project Geotechnical report)

Where specified on the drawings, the top layer of organic topsoil shall be stripped to the nominated depth from the area of the Works and stored for reuse or disposed of as directed by the Employer. Topsoil stockpiles shall not be more than two meters in height

unless otherwise approved. Sloped at 4 to 1 Horizontal to Vertical.

5.5 Excavations

5.5.1 Safety

The Contractor shall take the necessary actions to ensure the safety of each excavation with regard to people, equipment, existing services, nearby buildings, structures etc. Excavation, shoring and all activities associated with excavation shall be in accordance with Saudi laws and regulations and applicable referenced Standards.

The sides of all excavations shall conform to the following minimum requirements:

- Every face of any excavation that exceeds a depth of 1.5 m shall be supported or contained by shoring unless the face is cut back to a safe slope, or there is no danger to persons in the vicinity of the Work. The assessment of safety shall take into account the specific conditions of the excavation including soil characteristics, loading conditions and ground water levels. If deemed necessary or requested by the Employer the Contractor shall provide evidence of safety in the form of a stability analysis performed by a suitably qualified and experienced geotechnical engineer.
- The shoring of the excavation shall be braced as the excavation progresses and, where a mechanical digger is used, the shoring shall be kept as close as practicable to the digger.
- The shoring shall extend at least 200 mm above the surrounding ground level or a similar protection shall be provided at the edges of the excavation by a fender board.

Where excavations undercut, are next to, or reduce the support of structures, the structures shall be supported to the satisfaction of the Employer and to the extent necessary to ensure the safety and integrity of the structures.

All necessary measures shall be taken to ensure that excavations are in a safe condition at all times, including the erection of suitable safety barricades, warning signs and hazard lights.

5.5.2 General Requirements

The Contractor shall obtain permission in the form of a written permit from the Employer to carry out any excavation or penetration of the ground that is greater than 300 mm deep.

A permit to dig application shall be submitted for approval. Excavation shall not commence until approvals are obtained. The details of the proposed excavations shall be submitted to the Employer for approval at least three days prior to excavations commencing.

5.5.3 Dewatering

The Contractor shall be responsible for maintaining all excavations free from water, irrespective of its source, to the extent necessary for the execution of the Works or for safety reasons to the satisfaction of the Employer. The Contractor shall take all the action

and supply all equipment necessary to divert any surface and subsurface water from the Work. Trenches, berms, sumps, pumps, spears and other dewatering equipment shall be utilized as required.

The de-watering method and procedures shall not result in damage to or have any detrimental effect on adjacent structures and services or on adjoining land. The de-watering shall not result in excessive washout of fines from the de-watered area. The Contractor shall be responsible for all damages or detrimental effects caused by de-watering.

The Contractor shall take all necessary precautions at points of discharge of water to avoid causing excessive turbid run-off into stream courses, flooding or damaging the Works, adjoining Works, natural bush and property. Pollution of water courses is strictly prohibited. Should any accidental pollution occur, the Employer shall be informed immediately.

5.5.4 Excavation Depth

All excavations shall be made to the depth and extent shown on the drawings, with proper allowance for fill, blinding concrete, formwork and working space allowance, according to the actual Site conditions. The excavations shall be kept free and clean of loose materials, water and rubbish. Should excavation to the nominated depth reveal unstable or unsuitable ground, the Contractor shall immediately inform the Employer .

5.5.5 Over Excavation

Should the Contractor excavate to dimensions in excess of those specified, the Contractor shall, at his own expense, fill and compact the excess excavation with approved material and carry out additional trimmings as required. The Employer may require unauthorized over excavation to be made good using lean mix (15 MPa) concrete. Unless otherwise indicated in the drawing, Trench excavation shall be in accordance with PIP CVS02700.

5.5.6 Spoil

Excavated material that is unsuitable for backfill or other fill requirements, or is otherwise not required for re-use in the Works, shall be removed from the Site and disposed of at an approved Site. Stock piling of such material shall be done in accordance with referenced local regulations and standards.

5.5.7 Fill and Embankment Slopes

The slope of compacted fills or excavated embankment surfaces shall not be steeper than those shown on the drawings.

5.6 Rock Excavation

5.6.1 Classification of Rock Excavation

Rock excavations shall be determined as follows.

5.6.1.1 Bulk Rock Excavation

Bulk Rock excavation is that which would normally be carried out using a bulldozer and loader, excavator or scraper.

Rock shall be deemed to be material, which cannot be ripped by the machines noted in Table 5-1, that are in good condition, fitted with good tracks and single hydraulic ripper with a sharp ripper shoe; at the associated production rate (measured in-situ) in normal bulk earthwork excavation.

Table 5-1 : Reference Rock Ripping Machines

Machines	Production Rate Limit (m^3/h)
D11	135
D10T	105
D9R	90

5.6.1.2 *Detailed Rock Excavation*

This classification is intended for the situations where the space is limited or where a narrow trench is required. In such case, the material shall be deemed to be detailed Rock excavation if the material cannot be excavated at a rate of 200 m^3/h (measured in-situ) using a 110 kW flywheel power hydraulic excavator (equivalent to a Caterpillar E240C) in good condition and fitted with a 500 mm wide Rock bucket with sharp tines.

5.6.2 ***Testing Machinery***

The testing machinery for Rock excavation classification shall be provided together with a competent operator to demonstrate to the satisfaction of the Employer that any material warrants classification as Rock. Such material shall only be classified as Rock if it is actually removed using a Rock breaker, pneumatic pick or explosives.

5.7 Blasting

5.7.1 ***Safety***

All necessary measures are to be taken to ensure the protection of persons, the Work, adjacent plant, vehicles and property. To ensure protection from flying Rock, blasting nets shall be used where required for each blasting operation.

5.7.2 ***Regulations***

Where permitted, all drilling and blasting shall be performed strictly in accordance with the requirements of the Employer, Saudi laws and Regulations, and other statutory requirements as applicable.

5.7.3 ***Site Blasting***

If explosives will be needed for the Work on Site, the Contractor shall submit a written request to the Employer for approval to bring the explosives to Site.

This request shall include the following information:

- The details and location of the Work where explosives are proposed to be used

- The maximum quantity and types of explosives to be stored on Site
- The details of the explosives magazine/store and its proposed location
- Copies of current Shotfirer's Licences and permits.

Only after the Employer has given written permission shall the Contractor bring explosives on to the Site.

The Owner's Engineer shall be given seven days notice of any explosive storage requirements. Explosives shall only be stored in an approved magazine.

A Permit to Blast shall be obtained from the Employer prior to the Contractor carrying out any blasting. Written applications for a Permit to Blast shall be submitted to the Employer no later than 2 weeks before the planned blasting.

The application shall contain the following:

- The size, type and location of charges
- The method of protection
- The method of firing
- The time of firing requested
- The details of warning signals.

The excavation profile resulting from the blasting shall have a minimum of overbreak. The Employer shall be immediately notified of any blasting misfire, the location of the misfire and the remedial action taken.

All unused explosives shall be returned to the magazine immediately once the charge has been placed.

The exposed Rock faces/surfaces after blasting shall be made stable.

In special circumstances, the Employer may require blasting to be carried out by mine personnel. In these circumstances, the Contractor shall provide assistance in the form of labor, equipment and materials where required.

5.8 Filling

5.8.1 General Requirements

Unless notified otherwise, all the required filling materials shall be won, transported, conditioned, placed, compacted and tested by the Contractor. Where imported fill is required or specified, the fill shall be supplied by the Contractor.

Comment [F1]: Define. Purchased? Owned

5.8.2 Borrow Pits

The establishment of the borrow areas shall include clearing and grubbing the borrow pit site, removing and stockpiling topsoil, stripping and stockpiling any excess or unsuitable overburden, drainage and dust control, and, if required, a single access road or track. The

Contractor shall be responsible for, maintaining adequate bunting, barricading and signage to the borrow pit Site to prevent unauthorized entry to the area for the duration of the Works.

Comment [F2]: Define

The extraction of borrow material shall include excavation and processing of selected approved material. When processing of materials is required, it shall include combining with other materials; the addition of water and any other operation needed to produce the required material.

The Contractor shall operate and maintain borrow pits in accordance with the specified environmental limitations and restrictions and safety regulations. Borrow areas shall be kept free of large accumulations of water.

On completion of all operations at the borrow pit, the Contractor shall restore the pit as detailed below.

When the borrow pit is to remain exposed after the completion of the Work, the Contractor shall slope the sides of the borrow pit to a maximum slope of 1 vertical to 4 horizontal and erect appropriate barricading or signage to prevent unauthorized or inadvertent unsafe entry.

When the borrow pit is to be restored, the overburden and material in stockpile taken from the borrow area and not considered suitable for use in the Works, shall be spread evenly over the borrow pit area.

The finished profiles over the borrow area shall complement the natural contours and surface slopes of the surrounding ground.

The flow of water into the borrow pit shall be limited by the construction of earth banks and interception drains, together with any associated erosion protection Works as shown on the drawings.

Borrow pits shall be progressively established and rehabilitated in stages as soon as practicable.

5.8.3 Fill Material

The suitability of material for fill or backfill shall be approved before such fill is placed. Allowance shall be made for the use of excavated material where possible.

Fill material shall be that obtained from excavations, borrow pits, or from external sources by the Contractor. Fill material shall comply with the requirements of Section 4.3.

The Employer may deem excavated material as being unsuitable for use as fill, in which case this material shall be removed to spoil and replaced with approved fill material.

Approved fill material shall be used for Bedding and backfilling for underground pipe, conduits and duct banks Control Low Strength material (CLSM) or flowable fill may be substitute for either structural or general filling material, where approved by the

Employer.

The mix design shall be proportioned as per Section 4.3.3.

5.8.4 Inspection

Filling shall not be commenced nor shall the next layer be spread on newly compacted fill, until the area has been inspected and approved by the Employer.

5.8.5 Compaction Proof Rolling

After stripping the topsoil or excavating for placement of concrete fill, gravel areas, paving or road surfaces, all foundations shall be proof rolled. The proof roller shall be a 6-12 tonne drum roller travelling at a speed not exceeding that recommended by the roller manufacturer.

At least 4 passes of the roller shall be made to achieve the compaction requirements. Completed areas should be capable of withstanding proof rolling without excessive settlement. The proof rolling shall eliminate any soft or loose soil spots in the foundation and achieve the specified density. Unless unstable areas are confined to a discreet portion of the area being proof rolled, if unstable areas exceed 20% of the area being proof rolled, the whole area should be ripped, re-compacted and re-tested.

5.8.6 Placement of Earth Fill

Fill shall be placed in a systematic manner in horizontal layers of approximately equal thickness. The depth of each layer shall not exceed the depth for which satisfactory compaction can be guaranteed using the proposed plant and, in any case, shall not exceed 300 mm loose soil thickness, unless otherwise approved by the Employer. Unless otherwise specified in drawing. Bedding and covering for underground pipe shall be in accordance with PIP CVS02700. Cable and conduits bedding shall be placed in accordance with MD-101-SMEM-EG-EL-SPC-0001

No stone within the fill layer shall exceed one-third the compacted layer thickness in maximum dimension or 75 mm whatever is smaller. Screening or otherwise sorting of the material prior to placement may be required.

Where existing soil surfaces are to be covered with less than 200 mm of fill, the surface shall be scarified then mixed with the imported fill to give a minimum compacted layer thickness of 200 mm, to the same density as that specified for fill.

Areas to received fill that have soft or loose soil shall be compacted or removed and rejected with properly compacted fill. Unless otherwise permitted by the Employer, backfill around or over cast-in place concrete shall not be permitted until the concrete has attained 75% or its specified strength.

5.8.7 Moisture Conditioning

Prior to the compaction of fill, all fill material, whether from Site excavations or not, shall be moisture conditioned (if required), to achieve a moisture content within plus or minus two percentage of the optimum moisture value as determined by the applicable Proctor

test unless otherwise nominated in the project specific specification. The moisture shall be uniformly distributed throughout the fill and there shall be no clumps of soil. Sands shall be compacted either dry or full wet.

Unless stated otherwise, the supply of water for compaction purposes is the responsibility of the Contractor. The Contractor shall request approval of the proposed source of water supply. Request shall be submitted to the Employer within 48 hours of use and if source is no-potable the request shall include a water quality analysis.

5.9 Compaction of Earth Fill

5.9.1 Compacted Soil Density

Structural Fill shall be compacted to at least 95% of the Maximum dry Modified Proctor density, at a moisture content within two percentage points of the laboratory determined optimum value by ASTM D1557. Controlled Low-Strength Material (CLSM) may also be used (refer Section 4.3.3). Structural Fill that is granular material shall be compacted to 80% relative density, as determined by ASTM D4253 and ASTM D4254.

Fill other than Structural Fill shall be compacted to at least 90% of the maximum dry Modified Proctor density, as determined by ASTM D1557 Sand without a well-defined moisture-density curve shall be compacted to 70% relative density, as determined by ASTM D4253 and ASTM D4254.

The moisture content of the material being compacted shall be within ±2% of the optimum moisture content as determined by the applicable ASTM D1557. Dry or nearly dry sand can be compacted at their natural moisture content. The Contractor shall condition all fill materials as necessary to achieve the required moisture content during compaction.

Compaction by flooding is permitted as long as specified density is achieved.

Brackish water may be used for all compaction Work if approved by the Employer except as indicated below where potable water shall be used:

- Top layer of bulk backfill or structural backfill
- Aggregate base and sub-base courses for road construction
- All courses above Geomembranes, and
- Any fill within a distance of 1 m from all surfaces of structures/footings or utilities.

Sea water shall not be permitted to be used for any earthwork.

5.9.2 Backfill Compaction

Backfill around concrete and in trenches shall be compacted to a density at least as nominated in the drawings or project specification. Special care shall be taken when backfilling near retaining walls. Fill material shall not be placed behind the retaining wall until material on the low side has been compacted to specification (in layers, to density, at

right moisture etc.)

Fill and Backfill adjacent to structures such as retaining walls, pits, and basements shall not be compacted with heavy equipment , but with hand operated equipment to a distance of 4 feet (1.2m) or greater beyond the sides of the structures.

Heavy vibrating compaction equipment shall not be operated closer to the retaining wall than the height of the backfill on which it is operating. However, the compaction of the total fill volume must still comply with this specification.

Fill in trenches shall be compacted to a density at least equal to that of the surrounding in-situ material.

Backfill material behind retaining walls and similar structures shall be executed in a way that avoids damages to the membranes, waterproofing and/or drainage materials.

5.10 Finishing

The finished surface of all gravel areas, roadways and embankments, whether newly constructed or disturbed, shall be graded to a smooth surface.

Topsoil shall be spread on areas to be restored, as directed by the Employer or as shown on the drawings. The average slope of batters shall not exceed the specified slope.

The surface of the sub-grade under concrete or paving at any point shall not be higher than the design level shown on the drawings. Unless noted otherwise on the drawings, all levels shall be finished to the tolerances shown below.

Items	Tolerance
Sub-grade for concrete, paving, roadways and hardstand areas	+0 mm –20 mm
Detailed excavation: Deviation from the design level Deviation from a 3 m straight edge laid along the finished surface	 ±20 mm <10 mm
Bulk earthworks levels	±50 mm
Deviation of batter line from the design batter at any point along the batter	<150 mm
Horizontal Alignment	±50 mm in 100 m
Slope Alignment	±75 mm measured normal to slope

5.11 Dust Suppression

The Contractor shall be responsible for the control of all dust from all Construction areas for the duration of the Contract. This shall include all necessary watering of haul roads used for access and control of dust from stockpiles, borrow excavation and fill areas. The water source proposed to be used for dust control shall be agreed with the Employer. Use of water shall not result in, or create, hazardous or objectionable conditions.

The Contractor shall take all necessary measures to ensure that dust from the operations does not constitute a hazard or nuisance on the Works or adjacent areas. The Contractor shall comply with all local authority requirements and to the approval of the Employer.

6. TESTING

The Contractor shall submit to the Employer for approval an Inspection and Test Plan (ITP) which shall outline tests to be carried out to certify construction to the requirements of the specification, including hold points.

The Contractor shall engage an independent Registered Contractor for the purpose of carrying out laboratory and field testing of water, concrete, Rock, soils and aggregates.

The Contractor shall ensure all testing is completed in accordance with the approved ITP.

The laboratory used for soil and aggregates testing should be accredited by an appropriate governing body for material testing.

At least (1) control test shall be performed for gradation, density, plastic, and moistures content for each type of subgrade/fill material or where there is change in the type or source of material. These test include the following:

- Sieve Analysis (ASTM C136)
- Liquid and plastic limit Determination (ASTM D4318)
- Modified Proctor Test (ASTM D1557) or Standard Proctor Test (ASTM D698)

For granular soil that do not exhibit a well-defined moistures density relationship, index density, unit weight, and relative density shall be determined in accordance with ASTM D4253 and ASTM D4254.

As a minimum, one in-place moisture content and one density test shall be performed on every lift of fill. The number of field density tests shall be increased if a visual inspection determines that the moisture content is not uniform or if the compaction effort is variable and not considered sufficient to the specified dry density.

Field density testing shall be performed at the following frequencies:

- Structural fill under foundation and building slaps – one test every 1000 square feet (90 square m) of each lift.

- Structural fill under roadways, railroads, area pavement, and parking areas – one test every 2000 square feet (180 square m) of each lift.
- Road base and sub-base – one test every 2000 square feet (180 square m) of each lift.
- Backfill of trenches – one test every 50 linear feet (15 m) of each lift.
- General fill – one test every 5000 square feet (460 square m) pf each lift.

7. SUBMITTALS

The Contractor shall transmit the following submittals to the Employer:

- A Detailed Work Plan
- A Quality Control Program
- A statement attesting to compliance of the Contractor's Work with the requirements of this specification and other Contract Documents
- Site specific Inspection and Test Plan (ITP) and Test Reports for review and acceptance as mentioned in Section 6. The Inspection and Test Plans shall clearly identify participation of all parties concerned including but not limited to, the Contractor and Testing Agency. The ITP shall identify contract documents referenced to for activities to be undertaken, as well as all hold points
- Sources, and test results of all borrow materials and water used for fill
- Informal daily "pass" or "fail" reports, and non-compliance reports. Formal reports including all test logs and comments shall be submitted weekly. These formal reports shall include density and moisture content test logs, indicating location of tests by coordinates and elevation, and any comments. All test logs and comments shall be compiled and submitted at the completion of earthwork for permanent project records, and
- Accurately recorded as-built locations of project-installed buried utilities and tie-in points and interferences encountered during project construction. Data shall include horizontal dimensions, elevations, size and slope gradients, as appropriate for the utility.

8. QUALITY CONTROL

The Contractor shall refer to the following.

MD-101-SMPM-PM-QA-PEI-0001 Contractor Quality Requirements